When Self-Consistency Backfires: Majority Vote Hurts the Majority of Hard Science Problems for Small LLMs

Utkarsh Bahuguna
Scaler School of Technology
utkarshbahuguna10@gmail.com

Abstract

Self-consistency (SC) via majority vote is a widely used way to spend inference-time compute: sample N chains of thought, return the plurality answer. On the full GPQA Diamond benchmark (198 graduate-level science questions), majority voting reduces per-problem accuracy on a majority of problems for two instruction-tuned models from different families: 56.6% of problems for Qwen2.5-7B and 65.7% for Llama-3-8B, with Qwen the primary demonstration and Llama corroborating the direction from a near-chance baseline. The effect was pre-registered on a 151-problem confirmatory split after being observed on 47 exploratory problems, and all four confirmatory hypotheses passed. A grid oracle that routes each problem to the best N across $\{1, 2, 4, 8, 16, 32, 64\}$ marks a theoretical upper bound 14 accuracy points above $N = 1$ for Qwen and 17 for Llama, an oracle bound requiring ground truth rather than a deployable method. No verifier-free gate reaches it: a plurality-agreement gate captures 19% and 23% of that headroom on the pooled set, and neither it nor a token-entropy gate leaves accuracy statistically distinguishable from fixed-budget voting. The mechanism is direct: confidence does not track correctness on these problems. In the highest-agreement bin the plurality answer is correct about half the time for Qwen, and for Llama accuracy actually decreases with agreement. We pre-register and confirm these findings on small instruction-tuned models; we do not test reasoning-native models, which we flag as the central open question.

1 Introduction

Inference-time compute is a primary lever for LLM reasoning, and self-consistency—sample several chains of thought, return the plurality answer—is among the simplest and most widely used techniques in this family (Wang et al., 2023). It is commonly assumed to be a low-risk accuracy boost: sample more, vote, do at least as well.

That assumption fails on hard problems. When a model places its highest probability on an incorrect answer across independent samples, more samples only entrench the wrong vote. We call this *backfire*: $\text{mv\_gain} = \text{MV\_acc}(64) - \text{MV\_acc}(1) < 0$, where $\text{MV\_acc}(N)$ is the expected accuracy of a majority vote over N samples. We then ask the question a cost-conscious practitioner would ask: can a cheap, verifier-free signal computed from a few samples tell you which problems to vote on and which to skip? We test the two most natural signals, plurality agreement and token-level entropy, and find both fail, and we identify why.

That self-consistency can hurt is not itself new, and the underlying miscalibration is well documented (Guo et al., 2017; Kadavath et al., 2022). Our contribution is to quantify, pre-register, and confirm how often it happens on a full hard benchmark, across two model families, and to show that two cheap gates cannot prevent it.

Contributions:

- On all 198 GPQA Diamond problems, majority voting backfires on the majority of problems for two model families (56.6% and 65.7%), with Qwen the primary demonstration and Llama corroborating from a near-chance baseline. The effect was pre-registered on a 151-problem held-out split and confirmed (all four hypotheses passed).

- We show the per-problem routing headroom is real: a grid oracle (best N per problem across $\{1, 2, 4, 8, 16, 32, 64\}$) marks a 14 to 17 point ceiling above $N = 1$. A plurality-agreement gate captures 19 to 23% of that headroom on the pooled set. Neither it nor a token-entropy gate reaches the oracle or beats fixed-budget voting meaningfully.

- We give the mechanism: agreement does not track correctness, and for the weaker model it is anti-correlated.

2 Setup

Dataset and design. We use the full GPQA Diamond benchmark (Rein et al., 2023), 198 graduate-level multiple-choice questions in biology, chemistry, and physics. We adopt a pre-registered confirmatory design: 47 problems are **exploratory** (the hypotheses below were generated from them), and the remaining 151 are a **confirmatory** test set. The hypotheses and thresholds were locked and git-tagged before any confirmatory analysis. The repository, with tagged commits, is available at `https://github.com/u7k4rs6/compute-elasticity`. We report exploratory, confirmatory, and pooled (198) values throughout, but decide PASS/FAIL on the confirmatory set only.

Models. Two instruction-tuned models from different families, served on Together AI: Qwen2.5-7B-Instruct-Turbo and Meta-Llama-3-8B-Instruct-Lite. Both are small (7 to 8B) and non-reasoning. $N = 64$ samples per problem, temperature 0.7, single locked prompt template, byte-identical across all runs as verified by SHA-256. Five-pass answer extraction; parse rate 99.5% (Qwen) and 98.6% (Llama).

Metrics. $\text{MV\_acc}(N)$ is expected majority-vote accuracy over N samples, ties broken uniformly at random independent of ground truth, estimated by Monte Carlo over subsets. Backfire is $\text{mv\_gain} < 0$.

Routing and gates. The **oracle** routes each problem to the N that maximizes majority-vote accuracy across $\{1, 2, 4, 8, 16, 32, 64\}$ using ground truth; it is a theoretical upper bound, not a deployable method. The **agreement gate** returns the probe plurality when its fraction over k samples is at least τ, else votes at $N = 64$. The **entropy gate** routes by a threshold on mean per-token entropy. Both gates are verifier-free; ground truth is used only to score the result. Logprobs are available for all 151 confirmatory problems but not for the 47 older exploratory ones, so the entropy gate is evaluated on the confirmatory set. The entropy gate threshold is selected to maximize accuracy on the confirmatory set (Qwen: 0.113, Llama: 0.139).

Uncertainty. 95% confidence intervals are problem-level bootstrap (1000 iterations, seed 42).

3 Pre-Registered Confirmatory Results

All four pre-registered hypotheses pass on the 151 confirmatory problems (Table 1). Thresholds were fixed below the exploratory point estimates, so each is a genuine prediction rather than a restatement. PH2 and PH4 were pre-registered against a binary $\{N = 1, N = 64\}$ oracle; the grid oracle results in Sections 4.2 and 4.3 use a more generous ceiling and are reported separately as post-hoc analysis.

Hyp	Prediction (both models)	Qwen2.5-7B	Llama-3-8B	Result
PH1	backfire rate $\geq$ 33%	60.3% [53.0, 68.2]	65.6% [58.3, 73.5]	PASS
PH2	agree gate captures $\leq$ 10% of oracle	0.8%	-1.6%	PASS
PH3	top-agree-bin accuracy $\leq$ 70%	51.2% (n=43)	14.3% (n=21)	PASS
PH4	entropy gate captures $\leq$ 10% of oracle	0.5%	0.9%	PASS

Table 1: Confirmatory hypotheses ($n = 151$). Thresholds locked and git-tagged before analysis.

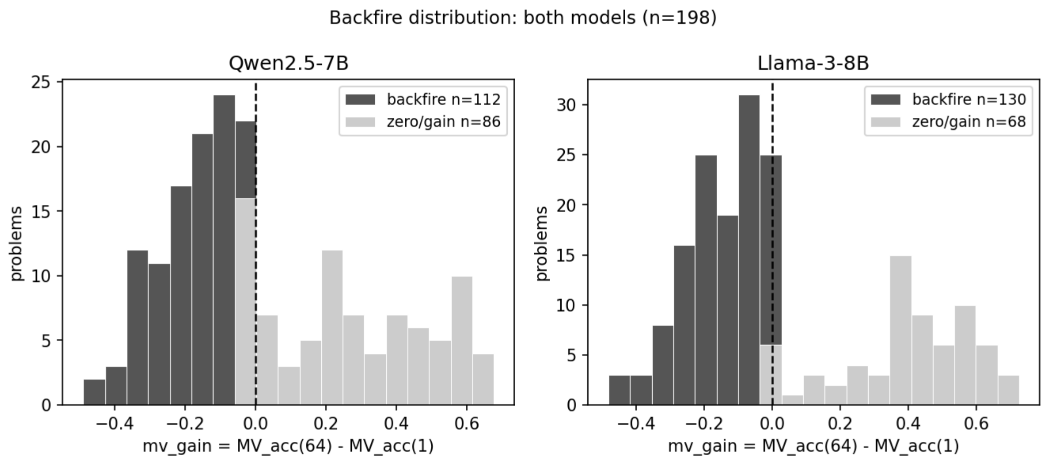

Figure 1: mv_gain $=$ MV_acc$(64) - MV_acc(1)$ per problem. Dark bars are backfire (mv_gain < 0).

4 Results (Pooled, 198 Problems)

4.1 Backfire affects the majority of problems, precisely estimated

Majority voting reduces per-problem accuracy on most problems for both models (Figure 1): the pooled backfire rate is 56.6% (95% CI [49.5, 63.6]) for Qwen and 65.7% ([59.1, 71.7]) for Llama. Expanding from 47 to 198 problems roughly halved the interval width (53% and 55% narrower), and both rates sit well above the 33% threshold.

The aggregate and per-problem pictures diverge. Voting barely moves aggregate accuracy (Qwen 0.342 to 0.369, Llama 0.273 to 0.313) while harming the majority of problems individually. An average that looks flat or mildly positive can hide widespread per-problem harm. The worst single problem loses 47 points (Qwen) or 46 (Llama); a few gain as much as 66 or 71. The asymmetry is real but rare: large gains exist, yet most problems backfire.

Backfire is not uniform across domains (Table 2). Chemistry is the dominant contributor: it accounts for 93 of the 198 problems and shows the highest backfire rate for both models (66.7% Qwen, 68.8% Llama). Physics is intermediate (50.0% Qwen, 67.4% Llama), and biology is least affected, though its small problem count ($n = 19$) makes that estimate noisy. We report this breakdown descriptively: the per-domain counts, biology especially, are too small to support claims about why backfire concentrates where it does.

4.2 The oracle upper bound is real but not reachable

A grid oracle that routes each problem to the best majority-vote accuracy across all N in $\{1, 2, 4, 8, 16, 32, 64\}$ reaches 0.482 (Qwen) and 0.439 (Llama), a ceiling 14 points above $N = 1$ for Qwen and 17 points above for Llama. This is a more generous bound than a binary

Domain	Qwen n	Qwen backfire	Llama n	Llama backfire
Biology	19	36.8%	19	42.1%
Chemistry	93	66.7%	93	68.8%
Physics	86	50.0%	86	67.4%
Pooled	198	56.6%	198	65.7%

Table 2: Per-domain backfire rate (pooled 198). Chemistry shows the highest backfire rate in both models and accounts for nearly half of all problems.

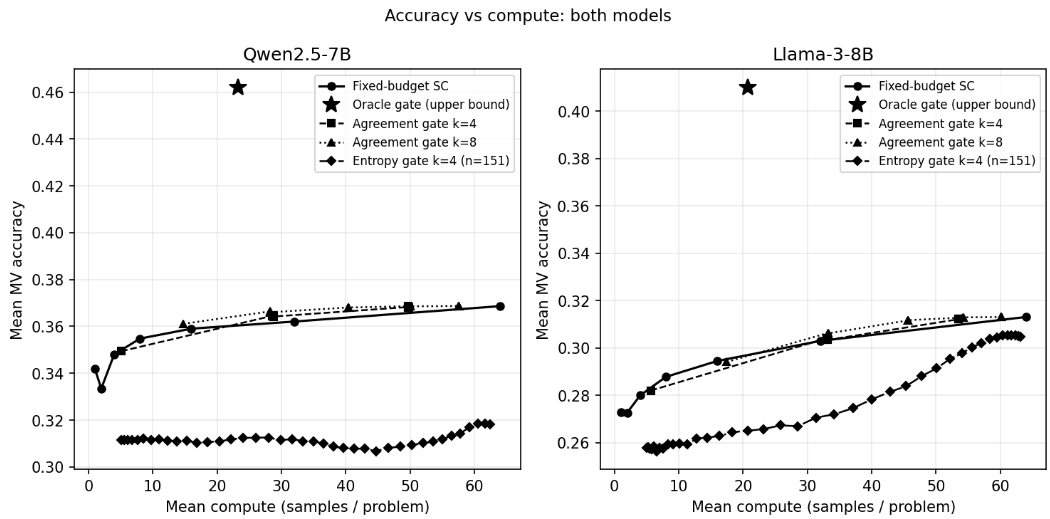

Figure 2: Accuracy vs mean compute: fixed-budget voting, the oracle upper bound, and the verifier-free agreement and entropy gates.

$\{N = 1, N = 64\}$ oracle, because it allows any compute level per problem. It marks how much accuracy a perfect per-problem routing decision could recover. It is an upper bound, not a method (Figure 2). The question this raises is how much of that ceiling is reachable without ground truth, which Section 4.3 answers by testing two verifier-free gates.

4.3 Two verifier-free gates fail to capture it

Neither cheap gate recovers the headroom. The agreement gate ($k = 8$, $\tau = 0.75$) reaches 0.368 accuracy for Qwen and 0.312 for Llama, statistically indistinguishable from fixed-budget voting at $N = 64$ (0.369 and 0.313). Against that ceiling, the agreement gate captures 18.7% of available headroom for Qwen and 23.4% for Llama (headroom is measured against the grid oracle of Section 4.2, not the binary oracle in Table 1): non-trivial fractions of a generous ceiling, yet accuracy remains flat. The gate spends k probe samples on every problem, then either returns the probe plurality (if the plurality fraction $\geq \tau$) or proceeds to $N = 64$; matched compute means the fixed-budget baseline is evaluated at the gate's realized mean samples-per-problem rather than a flat $N = 64$, so the comparison holds total inference cost equal. The entropy gate is evaluated entirely within the confirmatory 151, since logprobs exist only for that split. Measured consistently within it (the threshold was also selected on this set, so these are in-sample figures), the gate captures 0.2% of the grid-oracle headroom for Qwen and 31.2% for Llama, with gate accuracy of 0.318 (Qwen, threshold 0.113) and 0.306 (Llama, threshold 0.139). Because this capture is computed on a different problem set from the agreement-gate figures above, the two should not be read as like-for-like: the entropy percentages are confirmatory-151 values and the agreement percentages are pooled-198 values. In absolute terms the entropy gate is likewise statistically

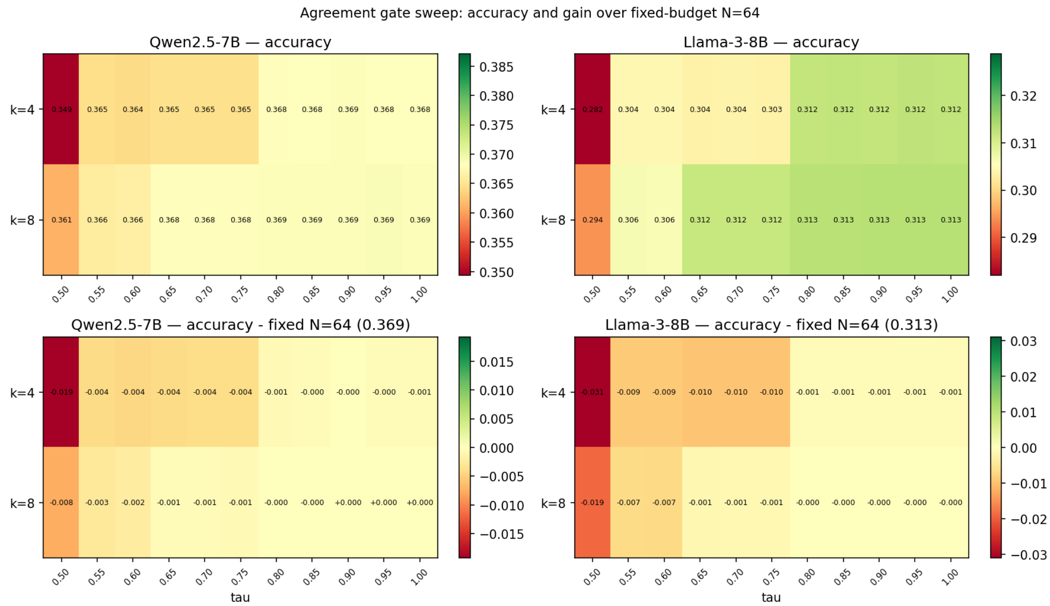

Figure 3: Agreement-gate accuracy across the (k, τ) sweep, relative to fixed-budget at matched compute. No operating point wins.

Confidence bin	Qwen n	Qwen frac correct	Llama n	Llama frac correct
[0.25, 0.50)	56	33.9%	79	30.4%
[0.50, 0.75)	83	27.7%	84	33.3%
[0.75, 1.00]	59	52.5%	35	28.6%

Table 3: Calibration by agreement bin (pooled 198). Confidence is the plurality fraction over all samples.

indistinguishable from fixed-budget voting. Mean per-token entropy modestly predicts which problems backfire (AUC 0.631 for Qwen, 0.523 for Llama), but predictive signal does not translate into routing accuracy because the gate cannot act on that signal without misclassifying enough problems to erase the gain. Across the full sweep of k and τ, no operating point beats fixed-budget meaningfully (Figure 3). The obvious agreement signal and an uncertainty signal both fail.

Localized entropy (computed over final-answer tokens only) was not computable from stored data, which retained only the mean scalar; this is a limitation of the pipeline rather than a finding.

4.4 Why: confidence does not track correctness

The reason is direct (Table 3, Figure 4). Binning problems by how concentrated the model's answers are, even the highest-agreement bin is far from reliable: Qwen's plurality is correct 52.5% of the time there, and Llama's only 28.6%, lower than its own low-agreement bin. Llama's accuracy decreases as agreement rises, an anti-calibration pattern. High self-consistency is therefore confidently wrong about as often as confidently right for Qwen, and more often for Llama. A gate that trusts agreement is reading a signal that does not carry the information it needs.

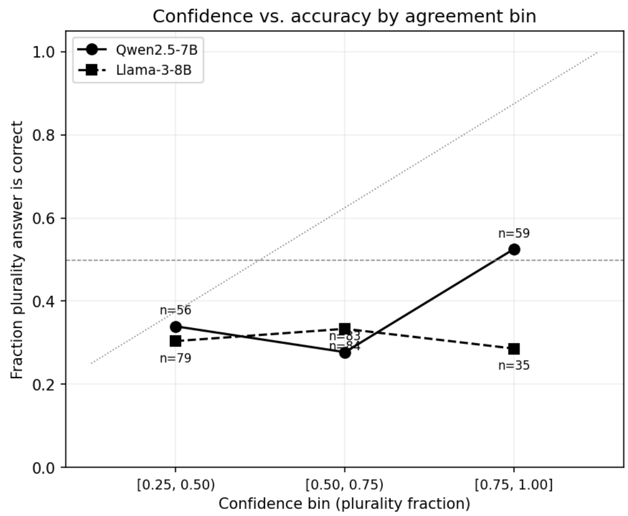

Figure 4: Fraction of plurality answers correct by agreement bin. Both models fall far below the calibration diagonal; Llama trends downward.

5 Discussion

Why backfire occurs. On hard problems the model's sampling distribution concentrates on a wrong answer, so voting locks in the error. Backfire is not a sampling artifact; it reflects the per-problem answer distribution. GPQA distractors are designed to be plausible (Rein et al., 2023), which amplifies the effect.

Why the gates fail. Both gates read confidence (agreement, or low entropy), but confidence on these problems does not indicate correctness, and for the weaker model is inversely related to it. The calibration data make this concrete.

Positioning. Prior work established that self-consistency helps on benchmarks where models have higher baseline accuracy (Wang et al., 2023); we show it hurts the majority of problems on a hard one. Repeated-sampling analyses study coverage under an oracle metric (Brown et al., 2024); we instead decompose the gap between realizable majority vote and the oracle. Adaptive-consistency early stopping (Aggarwal et al., 2023) is essentially our agreement gate, validated on easier datasets where backfire is rare; our negative result shows agreement stability is insufficient on hard problems. Closest to our work, Chen et al. (2024) show that majority-vote accuracy can rise and then fall as the number of LM calls grows, attribute this non-monotonicity to a mixture of easy and hard queries within a task (more calls help the easy ones and hurt the hard ones), and use that structure to estimate, from a small number of samples, the call count that maximizes aggregate performance. We differ in two ways. We measure the fraction of individual problems harmed rather than the shape of the aggregate curve, which matters because aggregate accuracy here looks nearly flat (0.342 to 0.369 for Qwen) while a majority of problems degrade underneath it. And our gate results sit in tension with few-sample estimation of an optimal call count: on a uniformly hard benchmark, the verifier-free signals such estimation would rely on do not recover the headroom. Tan et al. (2025) study self-consistent errors, where a model repeats the same wrong answer across samples, and find that consistency-based detectors, of which semantic cross-checking is one example (Zhang et al., 2023), fail on exactly those cases, which leads them to argue for an external verifier; their setting is error detection rather than compute allocation, but the conclusion converges with ours. The external-signal direction this motivates, trained verifiers and process reward models (Cobbe et al., 2021; Lightman et al., 2023), is the natural way to reach the oracle headroom, which our two verifier-free gates cannot.

Implications. On hard inputs, do not assume self-consistency is safe, and do not expect agreement or token entropy to tell you when it is; we tested both and both fail. Recovering the headroom likely requires a signal external to the model's own samples.

6 Limitations

Llama is near chance. Llama-3-8B's single-sample accuracy on full GPQA Diamond is 0.273, just above the 0.25 random baseline, so its backfire is less surprising. Qwen (0.342, clearly above chance) is the stronger demonstration; Llama corroborates the direction.

One benchmark. All results are on GPQA Diamond, graduate-level science. Other domains and easier difficulty regimes may differ.

Two small, non-reasoning models. Reasoning-native models require different infrastructure. We attempted a preliminary evaluation of a reasoning-native model with hidden chain-of-thought and found that a substantial fraction of samples exhausted the output token budget on hidden reasoning before producing a visible answer. This truncation made majority-vote accuracy incomparable to our non-reasoning models. A proper evaluation of reasoning-native architectures therefore requires larger inference budgets and explicit handling of hidden CoT output length, and is left to future work. Whether backfire shrinks or persists for such models remains the central open question.

Subset representativeness. The original 47-problem subset was easier than the full benchmark (Qwen $N = 1$ accuracy 0.418 vs 0.342 pooled), which is why we report on all 198.

Oracle and ratio. The oracle is an upper bound, not deployable, and the fraction-of-oracle-captured ratio is noisy (small denominator), so we lead gate claims with absolute accuracy.

Entropy threshold selection. The entropy threshold was chosen by an in-sample argmax over 40 candidate values on the confirmatory 151, maximizing capture against the binary $\{N = 1, N = 64\}$ oracle relative to an $\text{MV\_acc}(64)$ baseline, whereas the capture figures we report in Section 4.3 are measured against the grid oracle relative to $\text{MV\_acc}(1)$. Because selection and evaluation use the same problems, the entropy gate's reported capture is optimistic.

Limited gate family. We evaluate two verifier-free signals, plurality agreement and mean token entropy. Richer signals such as final-answer log-probability margins or confidence dynamics across samples could behave differently, and our negative result does not rule them out. Evaluating final-answer margins requires per-token log-probability arrays, which our pipeline did not store (it retained only a mean entropy scalar); this is left to future work.

Known mechanism. Confidence failing to track correctness is an instance of documented overconfidence (Guo et al., 2017; Kadavath et al., 2022); our contribution is the pre-registered, replicated, self-consistency-specific quantification and the failure of two cheap gates, not the existence of miscalibration.

7 Conclusion

On hard reasoning problems, self-consistency backfires on the majority of problems, pre-registered and confirmed on the full GPQA Diamond benchmark for Qwen and corroborated by a second family from a near-chance baseline. The accuracy this leaves on the table is real but unreachable with a verifier-free agreement or entropy gate, because confidence does not track correctness. Whether reasoning-native models escape this is the key open question.

References

Pranjal Aggarwal, Aman Madaan, Yiming Yang, and Mausam. Let's sample step by step: Adaptive-consistency for efficient reasoning and coding with llms. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 12375–12396, 2023.

Bradley Brown, Jordan Juravsky, Ryan Ehrlich, Ronald Clark, Quoc V Le, Christopher Ré, and Azalia Mirhoseini. Large language monkeys: Scaling inference compute with repeated sampling. *arXiv preprint arXiv:2407.21787*, 2024.

Lingjiao Chen, Jared Quincy Davis, Boris Hanin, Peter Bailis, Ion Stoica, Matei Zaharia, and James Zou. Are more LLM calls all you need? towards scaling laws of compound inference systems. In *Advances in Neural Information Processing Systems 37*, 2024.

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, et al. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pp. 1321–1330, 2017.

Saurav Kadavath et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let's verify step by step. *arXiv preprint arXiv:2305.20050*, 2023.

David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R Bowman. GPQA: A graduate-level Google-proof Q&A benchmark. *arXiv preprint arXiv:2311.12022*, 2023.

Hexiang Tan, Fei Sun, Sha Liu, Du Su, Qi Cao, Xin Chen, Jingang Wang, Xunliang Cai, Yuanzhuo Wang, Huawei Shen, and Xueqi Cheng. Too consistent to detect: A study of self-consistent errors in LLMs. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 4755–4765, 2025.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations*, 2023.

Jiaxin Zhang, Zhuohang Li, Kamalika Das, Bradley Malin, and Sricharan Kumar. SAC^3: Reliable hallucination detection in black-box language models via semantic-aware cross-check consistency. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 15445–15458, 2023.